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Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

Session	**2026-27 (ODD)**	**Course Name**	**PE-V : Lab : Blockchain Technology**
Semester	**7**	**Course Code**	**23IOT1748**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

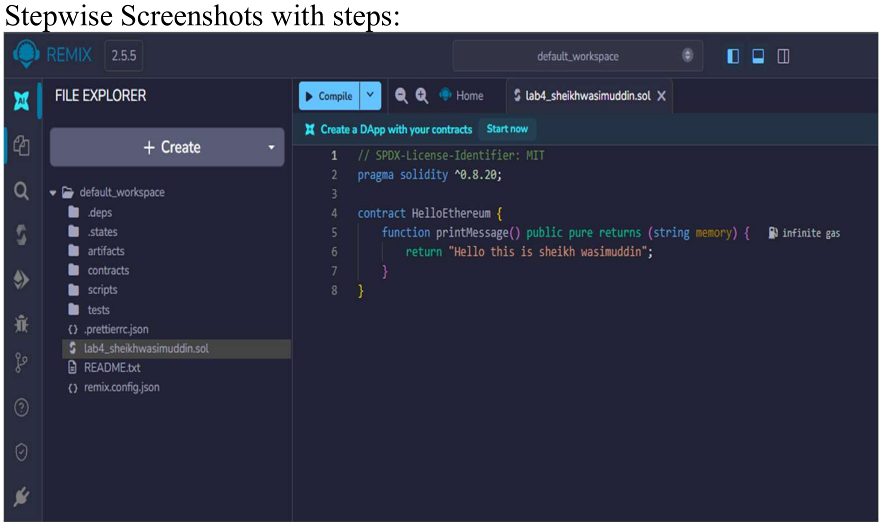
Practical Number	**04**
Course Outcome	**1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain.** **2. Outline various research aspects for development of blockchain.** **3. Make use of different security protocols to build secure blockchain applications.** **4. Analyze smart contracts and Hyperledger Fabric used in blockchain. 5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.**
Aim	Write and Execute Your First Solidity Smart Contract ("Hello Blockchain") Using Remix IDE.
Theory (100 words)	**1. Solidity** • Solidity is a **statically typed, contract-oriented programming language** mainly used for developing decentralized applications (DApps). • It is primarily used on the **Ethereum blockchain** and executes through the Ethereum Virtual Machine (EVM). • Solidity provides programming features such as **data types, functions, modifiers, structures, mappings, inheritance, and events**. • The Solidity source code is compiled into **EVM bytecode**, which can then be deployed to the blockchain. • It enables developers to create applications where the program logic is maintained and executed by the blockchain network. **2. Smart Contract** • A smart contract is a **blockchain-based program that contains predefined rules and instructions**. • It executes specific operations automatically when its functions are called and the required conditions are satisfied. • Each deployed smart contract has a **unique blockchain address** through which users and applications can interact with it. • Contract data and transactions are recorded on the blockchain,

	providing **transparency and traceability**. • Smart contracts are widely used in applications such as **digital payments, voting systems, token management, and decentralized finance**. **3. Remix IDE** • Remix IDE is an **online development platform for creating and testing Ethereum smart contracts**. • It provides tools for writing Solidity programs and checking them for compilation errors. • The **Solidity Compiler** in Remix converts source code into EVM-compatible bytecode. • The **Deploy & Run Transactions** environment allows contracts to be deployed and their functions to be tested. • Remix VM provides a **local blockchain simulation**, allowing experiments without using real Ether or a public blockchain network. **4. Hello Blockchain Contract** • The Hello Blockchain contract is a **basic Solidity application used to understand the structure and execution of a smart contract**. • It demonstrates how a contract is declared and how a function can return a predefined string. • The contract can be compiled using the Solidity Compiler available in Remix IDE. • After deployment on Remix VM, the function can be called through the deployed contract interface. • This example provides a basic understanding of the **Solidity → compilation → deployment → function execution** workflow.
Procedure and Execution (100 Words)	**Implementation Steps** 1. Launch a web browser and open **Remix IDE**. 2. From the **File Explorer**, create a new Solidity source file. 3. Save the file as **lab4_sheikhwasimuddin.sol**. 4. Open the created file in the Remix code editor. 5. Enter the following Solidity program: *// SPDX-License-Identifier: MIT* *pragma solidity ^0.8.20;* *contract HelloBlockchain {* *function printMessage() public pure returns (string memory) {*

```
        return “Hello this is sheikh wasimuddin";
    }
}
```

6. Select **Solidity Compiler** from the left navigation panel.
7. Choose a compiler version compatible with the program, such as **0.8.20**.
8. Click **Compile lab4_sheikhwasimuddin.sol.**
9. Check the compiler output and confirm that the contract is compiled successfully without errors.
10. Select **Deploy & Run Transactions** from the left sidebar.
11. Set the **Environment** to **Remix VM**.
12. Select **HelloBlockchain** from the **Contract** dropdown.
13. Set the transaction **Value** to **0 wei**.
14. Keep the **Gas Limit** set to **Auto**.
15. Click the **Deploy** button to deploy the contract.
16. After successful deployment, locate **Deployed Contracts** in the Remix interface.
17. Expand the deployed **HelloBlockchain** contract.
18. Locate the **printMessage** function.
19. Click the **printMessage** button to execute the function.
20. Observe the returned output: **Hello this is sheikh wasimuddin**
21.
21. Check the Remix terminal to observe the deployment and function execution details.
22. Verify that the smart contract has been **successfully compiled, deployed, and executed** in Remix VM.

Stepwise Screenshots with steps:



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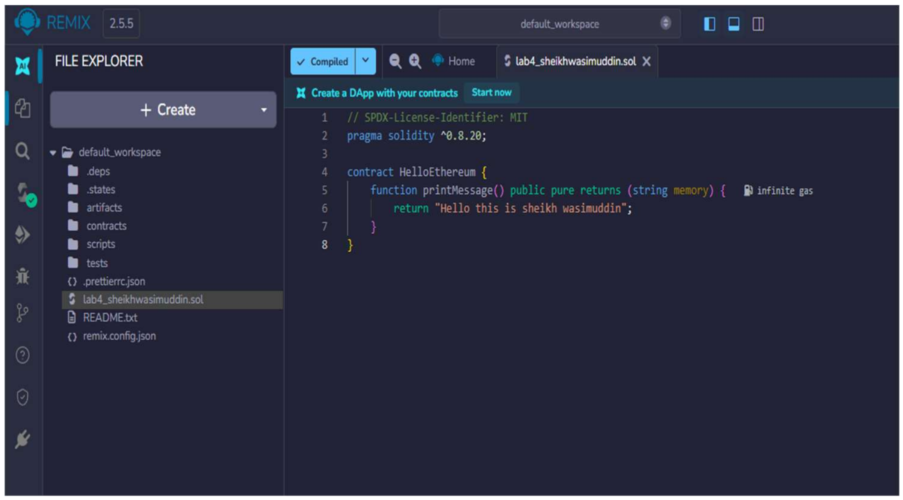

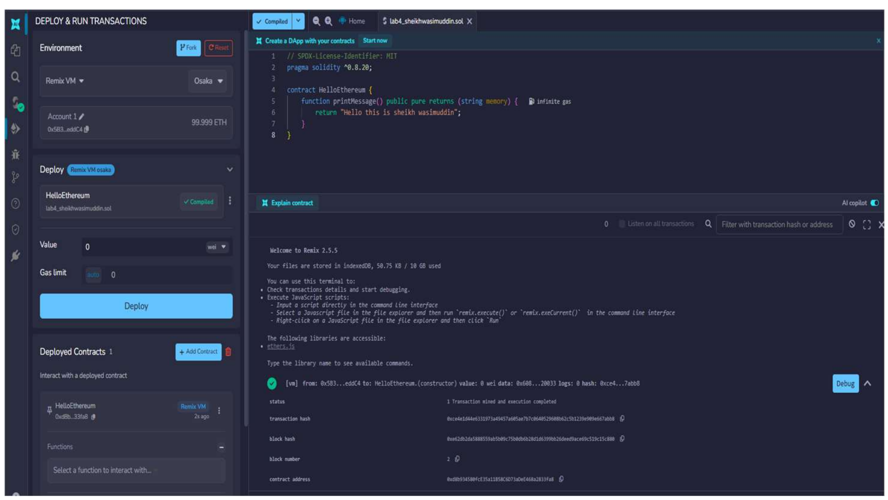

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	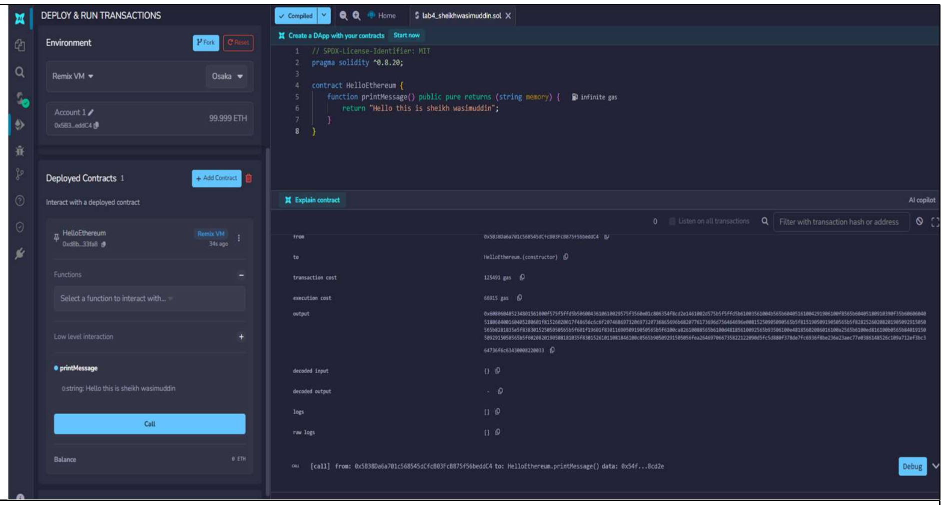
Conclusion	The practical successfully demonstrated the complete workflow of creating and executing a basic **Solidity smart contract** using **Remix IDE**. The HelloBlockchain contract was successfully written, compiled, and deployed in the **Remix VM** environment. The printMessage() function was executed successfully and displayed the output **"Hello this is sheikh wasimuddin"**. Hence, the practical provided a clear understanding of **Solidity syntax, smart contract compilation, deployment, and function execution** on a simulated Ethereum blockchain.
Date	25/08/2026